UniVersHOI: Unifying Interaction Versatility and Scene Scalability for Cooperative Human-Object Interaction

Anonymous CVPR submission

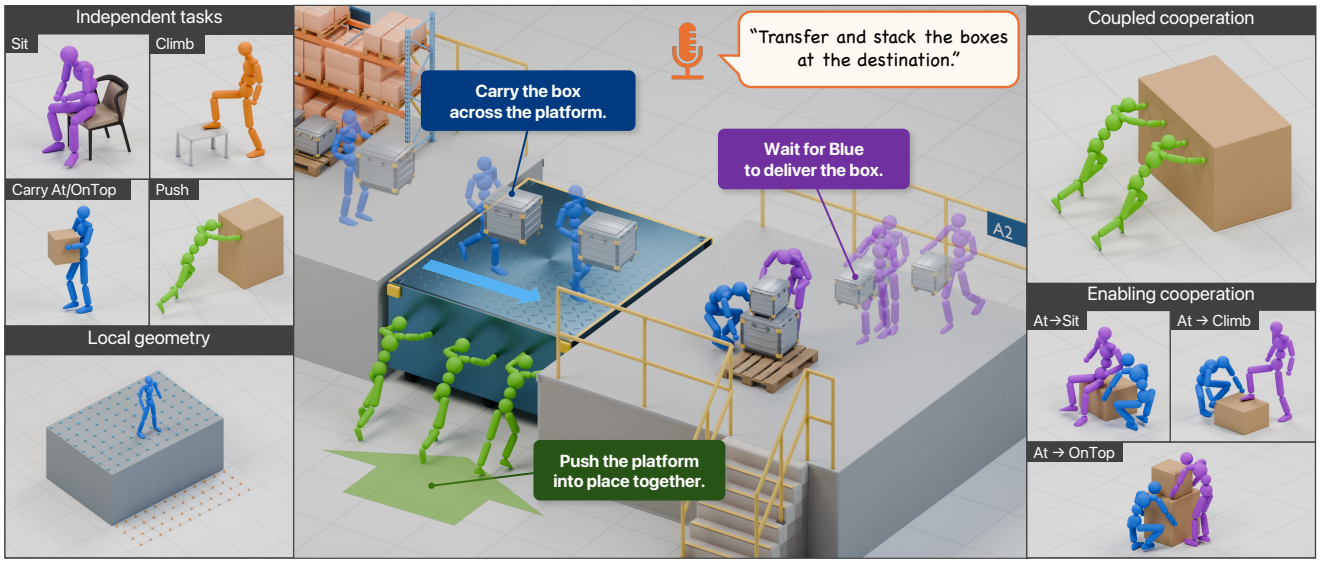

Figure 1. We present **UniVersHOI**, a unified physics-based policy that supports diverse interaction skills and inter-agent dependencies, including enabling cooperation between different tasks and coupled cooperation through joint object manipulation, across varying scene configurations. Combined with high-level task planning and local geometry observations, these capabilities enable the execution of complex cooperative scenarios.

Abstract

Multi-agent physical interaction requires humanoid agents to perform diverse interaction skills and coordinate under different inter-agent dependencies across varying scene configurations. Existing approaches primarily focus on versatile individual interactions or specific cooperative behaviors, leaving their unified control across diverse scenes challenging. We introduce UniVersHOI, a unified physics-based policy that combines interaction versatility and scene scalability for cooperative human-object interaction. Our framework consists of two complementary components. First, a Scalable Scene-Centric Transformer (SCT) jointly encodes humans, objects, and goals using task-relation attention biases and task-action fusion, enabling task-specific control across varying scene configurations. Second, a Dependency-Aware Coordination Transformer (DCT) incorporates inter-agent dependencies into human self-attention and feature representations, enabling coordination of independent, enabling, and coupled interactions without manually specified low-level coordination phases. The resulting policy generates actions for all agents in a single forward pass. We evaluate interaction versatility and scene scalability across varying numbers of agents, including configurations beyond the training scale, and analyze coordination under different dependencies. We further integrate an LLM planner and local geometry observations to execute cooperative scenarios from high-level instructions.

1. Introduction

When people work together, what one person can do often depends on what the others do. In Fig. 1, a group of simulated humanoids pushes a platform into place, others carry boxes across it, and one waits for a box to be delivered before stacking its own on top. Each agent performs a familiar interaction skill, but the scene succeeds only if each agent relates to the others in the right way: some must act in lockstep (*coupled*), some must wait for and build on a teammate's result (*enabling*), and the rest can proceed on their own (*independent*). Different combinations of these interactions can involve varying numbers and configurations of agents and objects, which a unified policy must accommodate.

Physics-based character control has advanced in both multi-skill interactions and cooperative manipulation, but largely along separate lines. Multi-skill methods give one humanoid a broad repertoire, such as sitting, climbing, and carrying, within a single policy [3, 24, 41]. Cooperative methods let several humanoids carry or push a shared object [2, 4, 19, 38, 42, 45], scaling to larger teams [19] and to sequential transport [9]. However, bringing these capabilities together in a single policy that handles diverse inter-agent dependencies remains largely unexplored. For example, existing work on sequential cooperation specifies execution phases and transitions through a hand-designed state machine [9], rather than letting a learned policy directly coordinate agents based on the evolving scene state.

We aim for interaction versatility: the ability to perform diverse skills under different inter-agent dependencies. Extending this versatility to scenes with varying numbers and configurations of agents and objects also requires scene scalability. Achieving both within a single policy is hard for two reasons. First, each agent must identify the entities relevant to its skill across varying scene configurations. Second, and more fundamentally, the policy must be versatile in how agents relate, not only in what each one does. How an agent should treat a teammate depends on the relation between their tasks, not just on either agent's skill: a climber must hold back while a teammate places the box it will climb, yet be ready the moment the box lands; agents pushing one platform must move in lockstep; and an agent on an independent task need not coordinate at all. Common workarounds fall short: per-agent multi-skill policies ignore these relations, so a climber may start before its box arrives. Even with hand-designed coordination procedures, handling changing situations requires additional rules, such as allowing a waiting agent to move out of a teammate's path.

We present UniVersHOI, a single physics-based policy that controls all agents in a scene and handles independent, enabling, and coupled interactions together. Our key idea is to treat task assignments and inter-agent dependencies alike as relations that condition attention. A Scalable Scene-Centric Transformer (SCT) encodes the scene as a variable set of agent, object, and goal tokens and conditions attention on task relations, in the manner of relation-aware transformers [27, 43], so that each agent focuses on the entities its skill involves. A Dependency-Aware Coordination Transformer (DCT) applies the same principle to inter-agent attention using dependency relations. Given task assignments and dependency relations, the policy determines when to act, when to wait, and when to act together based on the evolving scene state, without explicitly prescribed coordination phases.

We train the policy with reinforcement learning [26] and an adversarial motion prior [25], without cooperative motion demonstrations. We evaluate the same trained weights across varying scene sizes, including scenes that combine independent, enabling, and coupled interactions. Paired with an LLM planner and local geometry observations, the policy executes complex cooperative scenarios such as the one in Fig. 1.

Our main contributions are summarized as follows:

- We formulate cooperative multi-humanoid interaction through three relations (independent, enabling, and coupled) and present UniVersHOI, a single physics-based policy that handles all three without scripted coordination Phases.
- We introduce SCT to encode variable-sized scenes and condition attention on task relations, and DCT to condition inter-agent attention on dependency relations, enabling scalable multi-skill control and dependency-aware coordination.
- We show that learned dependency-aware coordination outperforms independent, cooperatively fine-tuned, and explicitly scheduled baselines in success rate, generalizes to scenes with 16 agents without retraining, and supports complex cooperative scenarios.

2. Related Work

2.1. Versatile Human-Object Interaction

Recent work on physics-based human-object interaction (HOI) has explored unified policies for diverse interaction tasks [29, 30, 32]. To achieve physically plausible interactions, PhysHOI [33] uses contact-graph rewards, while InterMimic [40] distills specialized imitation teachers into a unified controller. SkillMimic [34] learns reusable interaction skills, and SkillMimic-V2 [44] improves their robustness and transitions through demonstration augmentation. TokenHSI [24] takes a different approach, bringing heterogeneous skills together through composable task tokens. To make interaction control more flexible, other methods incorporate scene context [7], language-guided contact planning [37], and VLM-guided goal and reward design [3].

Generative interaction priors [41] and unified scene interaction and manipulation [12, 16] further extend these capabilities.While these methods offer versatile control for individual humanoids, extending such versatility to scenes with varying numbers of agents, objects, and goals poses a distinct challenge. Our method addresses this through a scalable scene representation with task-relation-conditioned attention, supporting reliable individual skill execution across multi-humanoid scenes.

2.2. Multi-Agent Interaction

Prior work on multi-agent interaction spans kinematic motion generation and physics-based character control. Kinematic approaches [5, 10, 17, 23] synthesize coordinated motions from language or observed partner motions. ReGenNet [39] captures asymmetric action-reaction dynamics, while InterControl [35] guides generation with pairwise joint-distance constraints. Physics-based studies [11, 31, 36, 45] investigate multi-character interactions in simulation. InterAgent [14] models fine-grained joint-to-joint spatial dependencies through interaction graphs, while AssistMimic [28] learns partner-aware policies for contact-rich assistive motion tracking. These approaches capture inter-agent coupling through body motion and physical contact. Our work instead focuses on dependency relations between tasks assigned to different humanoids, distinguishing independent, enabling, and coupled interactions. By explicitly modeling these relations, our policy coordinates diverse individual skills across varying multi-agent scenes.

2.3. Cooperative Human-Object Interaction

Cooperative motion modeling has advanced through interaction datasets and benchmarks [20], synchronized motion generation [8], contact-aware motion refinement [13], stability-driven motion generation with physical refinement [38], and interaction-preserving retargeting [18]. In physics-based control, Zhang et al. [45] reproduce cooperative carrying through interaction-graph-based imitation, while CooHOI [4] transfers individual interaction skills to cooperative control through shared object dynamics. TeamHOI [19] further supports varying team sizes through a decentralized policy with teammate tokens. Recent methods explore skill transfer and reuse in SynAgent [42] and From Solo to Ensemble [2], while Commander-Executor [9] combines high-level planning with physics-based control for cooperative object manipulation. Prior studies largely focus on shared-object manipulation, with some sequential approaches relying on hand-designed phases and transitions [9]. Our unified policy instead models dependency relations across assigned tasks to coordinate independent, enabling, and coupled interactions without prescribed coordination phases.

3. Method

3.1. Overview

Our goal is to develop a unified physics-based policy for multi-agent HOI that supports both interaction versatility and scene scalability. We denote the shared scene observation at time t by S_t, which contains information about objects, goals, and N humanoid agents. Task relations $\mathcal{R}_{\text{task}}$ specify each agent's interaction skill and its associated entities. Dependency relations $\mathcal{R}_{\text{dep}}$ describe dependencies among agents' assigned interactions, including enabling and coupled relationships. Given these inputs, a single policy generates physical actions for all N agents:

$$(a_{1,t}, \ldots, a_{N,t}) = \pi_\theta(S_t, \mathcal{R}_{\text{task}}, \mathcal{R}_{\text{dep}}), \tag{1}$$

where each action $a_{i,t}$ specifies joint targets for the corresponding agent. As illustrated in Fig. 2, the Scalable Scene-Centric Transformer (SCT) jointly encodes scene entities through entity self-attention, conditioned on task relations, to produce interaction-aware agent representations. The Dependency-Aware Coordination Transformer (DCT) then applies agent self-attention to these representations, conditioned on inter-agent dependencies. A shared action head maps the resulting representations to physical actions. We use proximal policy optimization (PPO) [26] with an adversarial motion prior (AMP) [25] for SCT pretraining and subsequent cooperative policy training.

3.2. Scalable Scene-Centric Transformer

Shared Scene Representation. Given S_t, we consider three entity types: agents, objects, and goals. Each entity u is represented by a feature vector $\mathbf{x}_u$ and a pose T_u. The pose maps the entity's local coordinate frame to a shared environment frame, specifying its position and orientation. Agent features encode articulated states, including body poses and velocities, while object features capture motion and bounding-box geometry. Goal entities use a constant feature, with target locations specified by their poses. Each feature vector $\mathbf{x}_u$ is mapped to a common embedding space by a type-specific MLP and combined with a learnable type embedding to form an entity token. Both the encoders and type embeddings are shared across entities of the same type.

The resulting tokens are jointly processed by a shared transformer through entity self-attention. To incorporate the relative positions and orientations of entities into attention, we adopt Geometric Transform Attention (GTA) [22]. Using entity poses T_u, GTA transforms queries, keys, and values into a shared coordinate space for attention, then maps the outputs back to each token's local frame. Detailed transformations are provided in the supplementary material. SCT produces representations for all agents in a single forward pass without separately encoding the scene for each agent. Since the encoders are shared within each

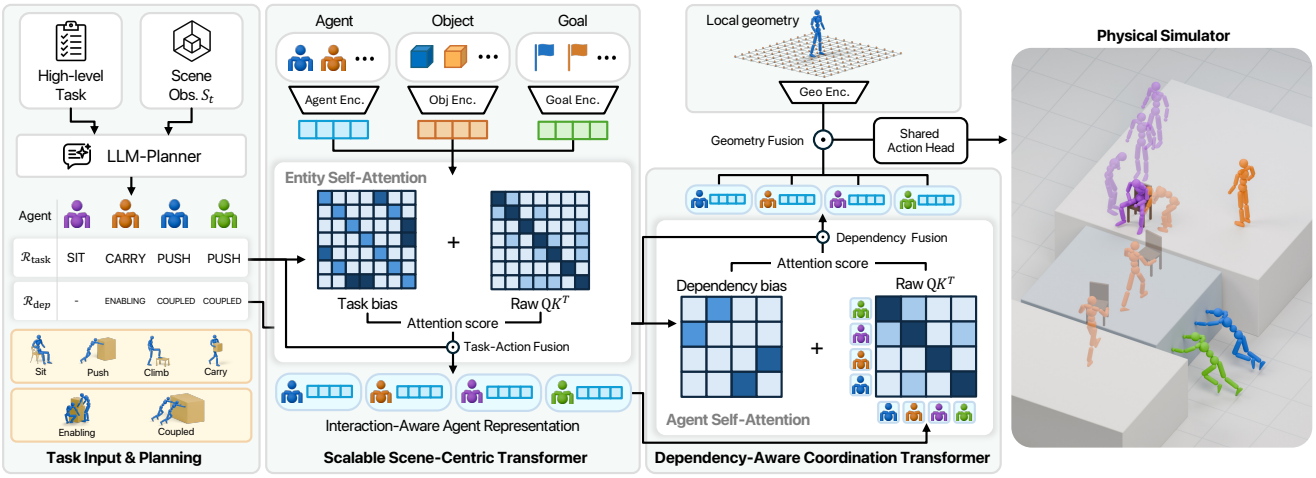

Figure 2. **Overview of UniVersHOI.** Given scene observations and task and dependency relations, the Scalable Scene-Centric Transformer (SCT) jointly encodes agents, objects, and goals using task-relation attention biases and task-action fusion. The Dependency-Aware Coordination Transformer (DCT) then incorporates inter-agent dependencies through dependency-relation attention biases and dependency fusion to generate coordinated actions through a shared action head. For complex cooperative scenarios, high-level task planning and local geometry observations further support policy execution.

entity type and the transformer supports variable-length inputs, SCT accommodates varying numbers of agents, objects, and goals without architectural changes.

Task Attention Bias. We consider five interaction skills: SIT, CLIMB, AT (carrying and placing an object at a target location), ONTOP (carrying and placing an object onto another object), and PUSH (pushing an object to a target location). Based on $\mathcal{R}_{\text{task}}$, we construct directed connections among tokens associated with each agent's assigned interaction. For example, SIT connects an agent token to its supporting object, while AT connects the agent to the manipulated object and goal, and the object to the goal.

For each task connection associated with skill s, we compute a scalar attention bias:

$$b_s = \mathbf{p}^\top \mathbf{e}_s. \tag{2}$$

Here, $\mathbf{e}_s$ is a learned skill embedding, and $\mathbf{p}$ is a projection vector learned separately for each transformer layer and attention head. These biases form $\mathbf{B}_{\text{task}}$, with zero bias for unconnected token pairs. The entity self-attention weights are

$$\mathbf{A} = \text{softmax}\left(\frac{\widetilde{\mathbf{Q}}\widetilde{\mathbf{K}}^\top}{\sqrt{d}} + \mathbf{B}_{\text{task}}\right), \tag{3}$$

where $\widetilde{\mathbf{Q}}$ and $\widetilde{\mathbf{K}}$ are GTA-transformed queries and keys, and d is the head dimension. All token pairs remain accessible through self-attention, including those without assigned task connections.

Task Fusion. While task attention biases guide information exchange among scene entities, task fusion directly conditions each agent's representation on its assigned skill. At each transformer layer, the post-attention feature $\mathbf{z}_i$ of agent i is updated through a residual MLP:

$$\mathbf{z}_i \leftarrow \mathbf{z}_i + f_{\text{task}}\left([\mathbf{z}_i; \mathbf{e}_{s_i}]\right). \tag{4}$$

Here, s_i denotes the skill assigned to agent i, and $[\cdot;\cdot]$ denotes concatenation. The skill embeddings are shared with the task attention bias, and f_{task} is shared across agents and transformer layers. The final agent-token outputs form the interaction-aware representations $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_N]^\top$, which serve as input to DCT.

We first train the scene-centric policy on the five interaction skills in two-agent scenes. Across all skills, task rewards comprise progress, state, and success terms, encouraging progress toward the objective, attainment of the desired physical state, and task completion, respectively. Detailed reward formulations and training settings are provided in the supplementary material.

3.3. Dependency-Aware Coordination Transformer

Dependency Attention Bias. Building on SCT's task-conditioned agent representations $\mathbf{H}$, DCT applies agent self-attention to model inter-agent dependencies specified by $\mathcal{R}_{\text{dep}}$. We consider two cooperative dependency types, ENABLING and COUPLED. ENABLING connects a follower's query to its predecessor's key, while COUPLED establishes bidirectional connections among agents jointly manipulating an object. Agents performing independent interactions have no assigned inter-agent dependencies.

For each dependency connection of type r, we compute a scalar attention bias from its learned dependency embed-

ding $\mathbf{e}_r$:

$$b_r = \mathbf{p}^\top \mathbf{e}_r. \tag{5}$$

Here, $\mathbf{p}$ is a DCT-specific projection vector learned separately for each layer and attention head. These biases form $\mathbf{B}_{\text{dep}}$ according to $\mathcal{R}_{\text{dep}}$, with zero bias for agent pairs without assigned dependencies. The agent self-attention weights are

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \mathbf{B}_{\text{dep}}\right), \tag{6}$$

where $\mathbf{Q}$ and $\mathbf{K}$ are queries and keys computed from agent representations. As in entity self-attention, dependency biases modulate attention without masking agent pairs.

Dependency Fusion. While dependency attention biases guide inter-agent information exchange, dependency fusion directly conditions agent attention contexts on dependency types. At each transformer layer, the attention context of each ENABLING follower or COUPLED participant is updated through a residual MLP:

$$\mathbf{m}_i \leftarrow \mathbf{m}_i + f_{\text{dep}}\left([\mathbf{m}_i; \mathbf{e}_r]\right). \tag{7}$$

Here, $\mathbf{m}_i$ denotes the attention-weighted value feature before the output projection, and the dependency embeddings are shared with the dependency attention bias. The MLP f_{dep} is shared across agents, dependency types, and transformer layers. Agents without applicable dependencies bypass dependency fusion. The DCT outputs form coordinated agent representations $\mathbf{C} = [\mathbf{c}_1, \ldots, \mathbf{c}_N]^\top$, which are mapped to physical actions through a shared action head.

Enabling and Coupled Cooperation. For enabling cooperation, a predecessor places an object at a target location (AT), allowing a follower to perform SIT, CLIMB, or ONTOP using the placed object. To support the follower's preparation, we use a virtual object token that retains the physical object's features and orientation but replaces its position with the predecessor's target position. The follower's task connections refer to this token, while physical interactions involve the actual object. For coupled cooperation, multiple agents jointly push a shared object. Each participant's PUSH task is connected to a separate object token representing the same physical rigid body. SCT processes these object tokens, while DCT models the corresponding inter-agent dependencies for joint manipulation.

Cooperative Policy Training. We separately train cooperative policies for enabling and coupled interactions in four-agent scenes. The pretrained SCT actor encoder is frozen, while DCT and the shared action head are optimized for each policy. We then distill the specialized policies into a unified cooperative policy supporting independent, enabling, and coupled interactions.

For enabling cooperation, the follower's progress reward remains active, while its state and success rewards are enabled only when the predecessor's AT objective is satisfied. Once the follower enters a waiting region, a movement penalty discourages further movement until the predecessor first satisfies its AT objective. For coupled PUSH, each agent is rewarded for approaching the shared object, while rewards for moving it toward the goal are enabled only when all participants are positioned to push. Detailed reward formulations and training settings are provided in the supplementary material.

3.4. Cooperative Scenario Execution

For cooperative scenario execution, we extend the controller with high-level task planning and local geometry observations. An LLM planner converts a high-level instruction and scene information into task relations $\mathcal{R}_{\text{task}}$ and dependency relations $\mathcal{R}_{\text{dep}}$ for the policy. While the planner determines the task structure, the learned policy generates physical actions without manually specified low-level coordination phases. Task and dependency assignments remain fixed throughout scenario execution.

To account for geometric constraints, we sample surface heights on an agent-centered, heading-aligned grid for each agent and encode the resulting height map using an MLP. The geometry feature is concatenated with the corresponding DCT representation and passed to a geometry-conditioned action head shared across agents. We freeze SCT and DCT and train the geometry encoder and action head for scenario execution. Detailed planner and geometry adaptation settings are provided in the supplementary material.

4. Experiments

We evaluate whether a single physics-based policy can perform diverse interaction skills under different inter-agent dependencies across varying scene configurations. We first assess interaction versatility and generalization beyond the training scale (Sec. 4.2), then analyze enabling and coupled coordination (Sec. 4.3). We examine the contributions of task and dependency fusion (Sec. 4.4) and demonstrate integrated execution in complex cooperative scenarios (Sec. 4.5).

4.1. Experimental Setup

Implementation Details. We implement our framework by extending the simulation environment provided by TokenHSI [24]. We use reference motions from AMASS [21], SAMP [6], OMOMO [15], and BONES-SEED [1] for AMP [25] and reference state initialization (RSI), without direct motion tracking or cooperative motion demonstrations.

Baselines. We adapt TokenHSI [24] to our five interaction skills and shared multi-agent setting, denoted as TokenHSI*. For multi-agent execution, TokenHSI* controls each agent independently using agent-specific observations

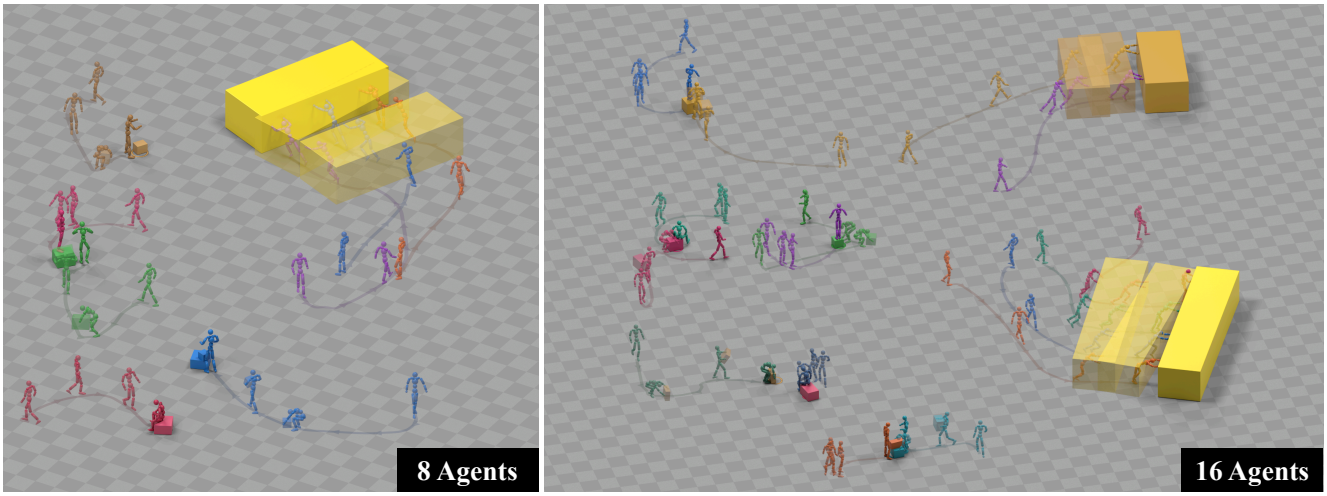

Figure 3. **Mixed-scene execution with 8 and 16 agents.** Independent, enabling, and coupled interactions coexist within each scene. Overlaid poses visualize execution over time, and lines trace the agents' actual movements rather than prescribed trajectories. In enabling interactions, overlapping or closely spaced poses of the follower indicate waiting while the predecessor completes the prerequisite task. In coupled interactions, agents that establish contact earlier wait for their teammates with little box displacement before pushing jointly toward the goal.

from the shared scene. We also compare against the pre-trained SCT and its cooperatively fine-tuned variant (SCT + Coop. FT). The latter replaces DCT with an MLP that aggregates features from partners and other agents, using the same input information and cooperative rewards as our full policy (Ours). Two additional baselines explicitly coordinate agent execution without further policy training: SCT + FSM (Finite-State Machine) uses state-based waiting and release rules, while SCT + LLM Scheduler estimates task-completion and approach times from initial scene observations to schedule agent start times. All methods use the same task assignments, with equivalent virtual-object information for enabling interactions and shared-object information for coupled interactions.

Evaluation Metrics. We measure task completion using success rate (SR) with task-specific spatial criteria. For AT and PUSH, success requires the physical object center to be within 20 cm of its target in 3D space, consistent with the carrying-task distance threshold in TokenHSI [24]. For SIT, CLIMB, and ONTOP, success requires the relevant point (pelvis for SIT/CLIMB and payload center for ONTOP) to lie within the supporting object's full horizontal extent and within 20 cm of the task-specific target height. We evaluate 512 scenes per setting per seed, with a maximum of 900 control steps (30 s) per episode, and report mean $\pm$ standard deviation across three evaluation seeds under matched evaluation conditions.

We compute SR at the interaction-group level: independent interactions are evaluated per agent, enabling pairs succeed when the predecessor's AT and follower's task conditions are simultaneously satisfied, and coupled groups succeed when their shared object reaches the goal. Each group contributes one outcome. For mixed scenes, we report the macro-average

$$\mathrm{SR}_{\mathrm{Mixed}} = \frac{\mathrm{SR}_I + \mathrm{SR}_E + \mathrm{SR}_C}{3}, \tag{8}$$

where SR_I, SR_E, and SR_C are computed over independent, enabling, and coupled groups within the mixed-scene evaluation, respectively. This gives equal weight to the three interaction types regardless of their sampling frequencies. Additional details on implementation, reference motions, training, baseline adaptations, and evaluation protocols are provided in the supplementary material.

4.2. Interaction Versatility and Scene Scalability

Tab. 1 evaluates Independent and Mixed SR across scenes with 2, 4, 8, and 16 agents. We evaluate each method with fixed model weights across all scene sizes, including 8- and 16-agent configurations beyond our four-agent cooperative training scale.

TokenHSI* applies the same multi-skill policy separately to each agent, rather than jointly controlling agents in a shared scene. Its Independent SR remains relatively stable across scene sizes, but its Mixed SR remains low, highlighting the limitations of per-agent control in handling inter-agent dependencies. In contrast, SCT jointly processes all agents, objects, and goals to generate actions for all agents through a unified policy. However, its limited Mixed SR shows that shared-scene modeling alone is insufficient for effective cooperative coordination.

Explicit coordination strategies, including SCT + FSM

Table 1. Interaction versatility and scene scalability. Independent (Ind.) and Mixed success rates (%) are reported as mean ± standard deviation across evaluation seeds.

	2H		4H		8H		16H	
Method	Ind.	Mixed	Ind.	Mixed	Ind.	Mixed	Ind.	Mixed
TokenHSI*	85.84±1.11	56.29±0.98	83.51±0.82	51.48±1.51	80.96±0.62	50.10±0.69	82.06±0.59	51.02±0.70
SCT	90.36±0.75	64.05±1.38	88.51±0.34	62.16±2.04	88.48±0.13	57.75±0.39	81.12±0.26	46.66±0.89
SCT + FSM	90.36±0.75	71.01±1.33	88.51±0.34	67.69±0.29	88.48±0.13	64.39±1.41	81.12±0.26	52.13±0.44
SCT + LLM Scheduler	90.36±0.75	68.54±3.13	88.51±0.34	64.61±2.36	88.48±0.13	57.25±2.47	81.12±0.26	49.69±0.62
SCT + Coop. FT	93.85 ± 0.39	79.27 ± 1.15	93.86 ± 0.60	75.27 ± 1.75	92.90 ± 0.61	71.98 ± 0.84	84.94 ± 0.27	59.37 ± 0.86
Ours	**97.40 ± 0.65**	**92.54 ± 0.95**	**96.55 ± 0.45**	**89.09 ± 0.97**	**94.87 ± 0.27**	**85.45 ± 1.06**	**90.15 ± 0.28**	**71.41 ± 0.27**

Table 2. Enabling cooperation. Group SR (%) for AT→SIT/CLIMB/ONTOP. Time (Succ./All) reports mean completion time over successful trials and all trials, respectively, assigning the maximum episode duration to failures. SR values are mean ± standard deviation across evaluation seeds; Time values are means.

Method	AT→SIT↑	AT→CLIMB↑	AT→ONTOP↑	Avg.↑	Time (Succ./All) (s)↓
TokenHSI*	14.19±0.30	5.73±0.30	11.26±0.69	10.39±0.19	9.90/27.91
SCT	19.01±1.66	35.29±1.30	21.09±1.28	25.13±0.69	12.29/25.60
SCT + FSM	45.77±2.52	43.88±0.56	28.91±3.58	39.52±0.81	14.79/23.99
SCT + LLM Scheduler	38.02±10.40	44.27±7.86	22.92±3.61	35.07±4.55	12.62/23.88
SCT + Coop. FT	63.09±1.47	54.36±2.99	52.86±3.16	56.77±1.08	12.20/19.90
Ours	**86.85**±2.44	**85.29**±1.77	**81.97**±0.81	**84.70**±0.79	10.64/**13.60**

Table 3. Coupled cooperation. Joint-PUSH Group SR (%) for two, three, and four participants, their average SR, and cooperative time ratio t_{coop} (%) adapted from TeamHOI [19]. Higher is better. Values are mean ± standard deviation across evaluation seeds.

Method	2H SR↑	3H SR↑	4H SR↑	Avg.↑	t_{coop} ↑
TokenHSI*	73.05±1.56	62.63±0.69	53.52±2.88	63.06±0.78	46.61±0.81
SCT	80.08±1.47	74.02±1.19	69.47±1.75	74.52±0.31	53.31±1.15
SCT + FSM	83.85±1.86	72.14±2.05	69.40±1.68	75.13±1.61	62.87±1.18
SCT + LLM Scheduler	73.44±2.71	66.67±3.93	59.90±3.25	66.67±1.38	44.64±1.99
SCT + Coop. FT	76.04±2.14	73.05±0.85	72.53±0.41	73.87±1.12	80.43±0.27
Ours	**95.18**±0.81	**92.19**±0.20	**87.57**±1.98	**91.64**±0.60	**84.05**

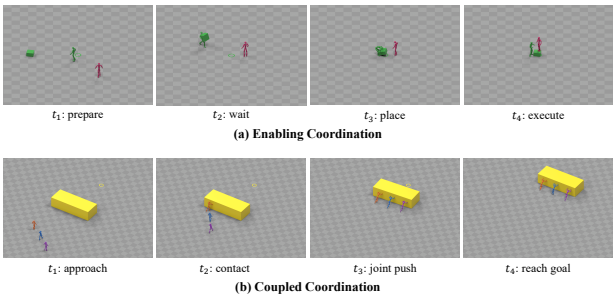

Figure 4. **Coordination behaviors in enabling and coupled interactions.** (a) While the predecessor carries a box to its goal, the follower autonomously waits until the box is placed, then climbs onto it. (b) Three agents approach a shared box simultaneously and establish contact. Agents that arrive earlier wait for their teammates; once all three agents establish contact, they jointly push the box to the goal.

and SCT + LLM Scheduler, introduce state-based waiting rules or precomputed execution schedules. Both methods improve Mixed SR over SCT but remain substantially below Ours, particularly in larger scenes. These results suggest that explicitly prescribed coordination strategies alone are insufficient to handle diverse inter-agent dependencies as scene complexity increases.

Cooperative fine-tuning improves SCT's Mixed SR by 12.71–15.22 percentage points across scene sizes. Nevertheless, Ours further outperforms SCT + Coop. FT by 12.04–13.82 points, despite using the same input information and cooperative rewards. This supports the benefit of DCT's structured dependency-aware coordination over MLP-based cooperative feature aggregation.

Overall, Ours achieves the highest Independent and Mixed SR across all evaluated scene sizes. UniVersHOI jointly processes the shared scene and generates actions for all agents in a single forward pass. Without additional fine-tuning, it achieves Independent/Mixed SR of 94.87/85.45% and 90.15/71.41% in 8- and 16-agent scenes, respectively. Although performance decreases in larger scenes, Ours maintains a substantial advantage over competing methods. Fig. 3 illustrates the concurrent execution of diverse interactions under the same unified policy.

4.3. Cooperative Interaction Performance

Enabling cooperation. Enabling interactions require the follower to wait for the predecessor's task completion while remaining ready to perform its dependent task. As shown in Tab. 2, TokenHSI* and SCT achieve low Group SR, while explicit waiting rules or execution schedules improve performance but remain limited. Cooperative fine-tuning increases the average Group SR to 56.77%, whereas Ours achieves 84.70%, a further improvement of approximately 28 percentage points. These gains are consistent across all three dependent skills: AT→SIT, AT→CLIMB, and AT→ONTOP. Ours also reduces the average completion time over all trials from 19.90 s for SCT + Coop. FT to 13.60 s, including failures. As illustrated in Fig. 4(a), the follower prepares and waits near the target location while the predecessor transports the object, then executes its de-

Table 4. Feature fusion ablation. Independent and Mixed SR (%) are evaluated for SCT and DCT, respectively. Ablated variants are retrained under matched training settings. Values are mean ± standard deviation across evaluation seeds.

Evaluation	Method	2H SR↑	4H SR↑	8H SR↑	16H SR↑
Independent	w/o Task Fusion	66.73±1.75	59.91±0.79	33.67±0.45	8.05±0.26
	SCT	**90.36**±0.75	**88.51**±0.34	**87.30**±0.66	**71.28**±0.67
Mixed	w/o Dep. Fusion	85.36±1.17	81.51±0.24	78.05±0.66	59.18±0.87
	Ours	**90.27**±0.77	**86.97**±0.87	**84.66**±0.87	**71.62**±0.64

pendent task after placement. This demonstrates coordination beyond explicitly prescribed task ordering, allowing the follower to prepare and act according to the predecessor's progress.

Coupled cooperation. Coupled interactions require agents not only to establish simultaneous contact but also to coordinate their actions for successful joint manipulation. Tab. 3 evaluates Group SR and the cooperative time ratio t_{coop} for joint PUSH with two, three, and four participants. Explicit execution control provides limited improvements in joint manipulation success, while SCT + Coop. FT increases t_{coop} from 53.31% for SCT to 80.43% without improving Group SR. This shows that sustained joint contact alone is insufficient for successful manipulation. In contrast, Ours achieves Group SR of 95.18%, 92.19%, and 87.57% for two, three, and four participants, respectively, averaging 91.64%, approximately 18 percentage points above SCT + Coop. FT. It also achieves the highest t_{coop} of 84.05%. As illustrated in Fig. 4(b), agents arriving earlier wait for their teammates, and all participants jointly push the object toward its goal after establishing contact. These results demonstrate that UniVersHOI achieves both sustained joint contact and successful manipulation, supporting distinct coordination behaviors within a single policy.

4.4. Ablation Study

Tab. 4 examines the contributions of task and dependency fusion. We retrain each ablated variant while retaining the corresponding attention biases and otherwise matching its training settings.

Task fusion. Removing task-action fusion reduces SCT's Independent SR across all evaluated scene sizes, with more pronounced degradation in larger scenes. Task-relation attention biases remain available to guide information exchange among entities. The additional gain from fusion therefore supports directly conditioning agent representations on their assigned skills alongside task-conditioned attention.

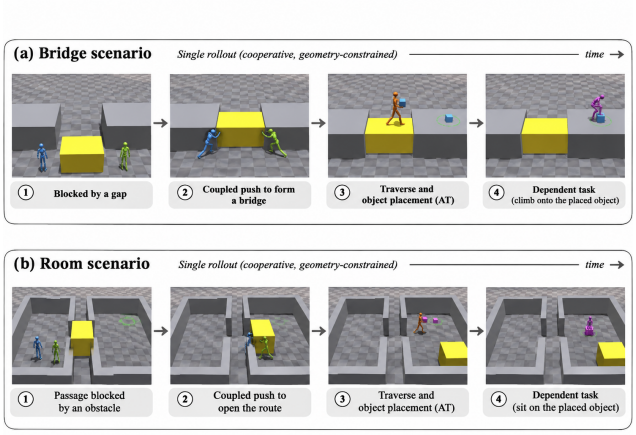

Figure 5. Cooperative scenario execution. (a) Bridge: joint PUSH establishes a traversable bridge, followed by object delivery and CLIMB. (b) Room: joint PUSH clears a passage, followed by object delivery and SIT.

Dependency fusion. Removing dependency fusion reduces Mixed SR at every scene size, with a larger gap in larger scenes. Since agent attention and dependency-relation biases are retained, this comparison supports directly conditioning attention contexts on dependency types in addition to modeling relations through attention. Together, these ablations show that explicit skill- and dependency-conditioned feature updates contribute to control across varying scene configurations.

4.5. Cooperative Scenario Execution

Fig. 5 demonstrates integrated execution in two four-agent scenarios. In Bridge, two agents jointly position a platform to establish a traversable path for object delivery and a subsequent dependent interaction. In Room, joint pushing clears a blocked passage, allowing the remaining agents to access objects and complete their assigned interactions. Both scenarios connect coupled manipulation with enabling interactions through changes to the physical environment.

An LLM planner supplies task and dependency assignments, which remain fixed throughout execution. To account for geometric constraints, we freeze SCT and DCT and train the geometry encoder and action head for these scenarios. The resulting policy coordinates physical execution without manually specified low-level coordination phases, demonstrating how the interaction capabilities evaluated above can be integrated with task planning and local geometry observations.

5. Conclusion

We presented UniVersHOI, a unified physics-based policy for cooperative human-object interaction that supports diverse interaction skills and inter-agent dependen-

cies across varying scene configurations. The Scalable Scene-Centric Transformer models task-relevant relationships among agents, objects, and goals, while the Dependency-Aware Coordination Transformer enables coordination according to inter-agent dependencies. Together, these components generate actions for all agents in a single forward pass, supporting independent, enabling, and coupled interactions without manually specified low-level coordination phases. We evaluate interaction versatility and scene scalability across varying scene sizes, including configurations beyond the cooperative training scale, and examine enabling and coupled coordination. Cooperative scenarios further demonstrate the integration of joint manipulation and dependent interactions with local geometry information. Extending the framework to broader forms of cooperative manipulation beyond joint pushing remains a direction for future work.

LLM Usage Statement

OpenAI

References

[1] Bones Studio. BONES-SEED: Skeletal everyday embodiment dataset. https://huggingface.co/datasets/bones-studio/seed, 2026. 5

[2] Zekai Deng, Kangyi Chen, Ye Shi, and Jingya Wang. From solo to ensemble: A hierarchical framework for composable multi-agent human-object interaction. *arXiv preprint arXiv:2610.11722*, 2026. 2, 3

[3] Zekai Deng, Ye Shi, Kaiyang Ji, Lan Xu, Shaoli Huang, and Jingya Wang. Human-object interaction via automatically designed VLM-guided motion policy. In *International Conference on Learning Representations (ICLR)*, 2026. 2

[4] Jiawei Gao, Ziqin Wang, Zeqi Xiao, Jingbo Wang, Tai Wang, Jinkun Cao, Xiaolin Hu, Si Liu, Jifeng Dai, and Jiangmiao Pang. CooHOI: Learning cooperative human-object interaction with manipulated object dynamics. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 79741–79763, 2024. 2, 3

[5] Anindita Ghosh, Rishabh Dabral, Vladislav Golyanik, Christian Theobalt, and Philipp Slusallek. ReMoS: 3D motion-conditioned reaction synthesis for two-person interactions. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[6] Mohamed Hassan, Duygu Ceylan, Ruben Villegas, Jun Saito, Jimei Yang, Yi Zhou, and Michael J. Black. Stochastic scene-aware motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 5

[7] Mohamed Hassan, Yunrong Guo, Tingwu Wang, Michael J. Black, Sanja Fidler, and Xue Bin Peng. Synthesizing physical character-scene interactions. In *ACM SIGGRAPH 2023 Conference Proceedings*, New York, NY, USA, 2023. Association for Computing Machinery. 2

[8] Wenkun He, Yun Liu, Ruitao Liu, and Li Yi. SyncDiff: Synchronized motion diffusion for multi-body human-object interaction synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11731–11743, 2025. 3

[9] Buzhen Huang, Xingchen Wu, Yuanwang Yang, Chenyi Xie, and Kun Li. Scalable multi-human cooperative manipulation via commander-executor coordination. *arXiv preprint arXiv:2610.14911*, 2026. 2, 3

[10] Muhammad Gohar Javed, Chuan Guo, Li Cheng, and Xingyu Li. InterMask: 3D human interaction generation via collaborative masked modeling. In *International Conference on Learning Representations (ICLR)*, pages 45683–45704, 2025. 3

[11] Kaiyang Ji, Ye Shi, Zichen Jin, Kangyi Chen, Lan Xu, Yuexin Ma, Jingyi Yu, and Jingya Wang. Towards immersive Human-X interaction: A real-time framework for physically plausible motion synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10173–10183, 2025. 3

[12] Jeonghwan Kim, Hyeonwoo Kim, and Hanbyul Joo. Co^{2}Skill: Whole-body control via skill composition for long-horizon human-environment interaction. *arXiv preprint arXiv:2610.09291*, 2026. 3

[13] Jiye Lee, Yonghun Choi, and Jungdam Won. MOCHI: Motion enhancement of collaborative human-object interactions. *ACM Transactions on Graphics*, 45(4):1–18, 2026. 3

[14] Bin Li, Ruichi Zhang, Han Liang, Jingyan Zhang, Juze Zhang, Xin Chen, Lan Xu, Jingyi Yu, and Jingya Wang. InterAgent: Physics-based multi-agent command execution via diffusion on interaction graphs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15253–15265, 2026. 3

[15] Jiaman Li, Jiajun Wu, and C. Karen Liu. Object motion guided human motion synthesis. *ACM Transactions on Graphics*, 42(6), 2023. 5

[16] Jianan Li, Xiao Chen, and Tien-Tsin Wong. Unifying physics-based humanoid interaction with a context-conditioned interaction prior. In *SIGGRAPH Asia 2026 Conference Papers*, 2026. 3

[17] Han Liang, Wenqian Zhang, Wenxuan Li, Jingyi Yu, and Lan Xu. InterGen: Diffusion-based multi-human motion generation under complex interactions. *International Journal of Computer Vision*, 132(9):3463–3483, 2024. 3

[18] Ziyu Lin, Shihang Du, Tianhang Wang, Xiaodong Liu, Runze Zhang, Zongyou Yu, Haodong Zhang, Alois Knoll, and Guang Chen. CoCarry: Contact-support-structured retargeting for dual-humanoid cooperative carrying, 2026. arXiv:2610.32460. 3

[19] Stefan Lionar and Gim Hee Lee. TeamHOI: Learning a unified policy for cooperative human-object interactions with any team size. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 37121–37132, 2026. 2, 3, 7

[20] Yun Liu, Chengwen Zhang, Ruofan Xing, Bingda Tang, Bowen Yang, and Li Yi. CORE4D: A 4D human-object-human interaction dataset for collaborative object REarrangement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1769–1782, 2025. 3

[21] Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael J. Black. AMASS: Archive of motion capture as surface shapes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5442–5451, 2019. 5

[22] Takeru Miyato, Bernhard Jaeger, Max Welling, and Andreas Geiger. GTA: A geometry-aware attention mechanism for multi-view transformers. In *International Conference on Learning Representations*, 2024. 3

[23] Sakuya Ota, Qing Yu, Kent Fujiwara, Satoshi Ikehata, and Ikuro Sato. PINO: Person-interaction noise optimization for long-duration and customizable motion generation of arbitrary-sized groups. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10676–10685, 2025. 3

[24] Liang Pan, Zeshi Yang, Zhiyang Dou, Wenjia Wang, Buzhen Huang, Bo Dai, Taku Komura, and Jingbo Wang. To-

kenHSI: Unified synthesis of physical human-scene interactions through task tokenization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5379–5391, 2025. 2, 5, 6

[25] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. AMP: Adversarial motion priors for stylized physics-based character control. *ACM Transactions on Graphics*, 40(4):144:1–144:20, 2021. 2, 3, 5

[26] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. 2, 3

[27] Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-attention with relative position representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 464–468, 2018. 2

[28] Yuto Shibata, Kashu Yamazaki, Lalit Jayanti, Yoshimitsu Aoki, Mariko Isogawa, and Katerina Fragkiadaki. Learning to assist: Physics-grounded human-human control via multi-agent reinforcement learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 3

[29] Chen Tessler, Yunrong Guo, Ofir Nabati, Gal Chechik, and Xue Bin Peng. MaskedMimic: Unified physics-based character control through masked motion inpainting. *ACM Transactions on Graphics*, 43(6):1–21, 2024. 2

[30] Guy Tevet, Sigal Raab, Setareh Cohan, Daniele Reda, Zhengyi Luo, Xue Bin Peng, Amit Haim Bermano, and Michiel van de Panne. CLoSD: Closing the loop between simulation and diffusion for multi-task character control. In *The Thirteenth International Conference on Learning Representations*, 2025. 2

[31] Jiashun Wang, Jessica Hodgins, and Jungdam Won. Strategy and skill learning for physics-based table tennis animation. In *ACM SIGGRAPH 2024 Conference Papers*, pages 1–11, 2024. 3

[32] Wenjia Wang, Liang Pan, Zhiyang Dou, Jidong Mei, Zhouyingcheng Liao, Yuke Lou, Yifan Wu, Lei Yang, Jingbo Wang, and Taku Komura. SIMS: Simulating stylized human-scene interactions with retrieval-augmented script generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14117–14127, 2025. 2

[33] Yinhuai Wang, Jing Lin, Ailing Zeng, Zhengyi Luo, Jian Zhang, and Lei Zhang. Physhoi: Physics-based imitation of dynamic human-object interaction. *arXiv preprint arXiv:2312.04393*, 2023. 2

[34] Yinhuai Wang, Qihan Zhao, Runyi Yu, Hok Wai Tsui, Ailing Zeng, Jing Lin, Zhengyi Luo, Jiwen Yu, Xiu Li, Qifeng Chen, Jian Zhang, Lei Zhang, and Ping Tan. Skillmimic: Learning basketball interaction skills from demonstrations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17540–17549, 2025. 2

[35] Zhenzhi Wang, Jingbo Wang, Yixuan Li, Dahua Lin, and Bo Dai. InterControl: Zero-shot human interaction generation by controlling every joint. In *Advances in Neural Information Processing Systems*, pages 105397–105424, 2024. 3

[36] Jungdam Won, Deepak Gopinath, and Jessica Hodgins. Control strategies for physically simulated characters performing two-player competitive sports. *ACM Transactions on Graphics*, 40(4):1–11, 2021. 3

[37] Zeqi Xiao, Tai Wang, Jingbo Wang, Jinkun Cao, Wenwei Zhang, Bo Dai, Dahua Lin, and Jiangmiao Pang. Unified human-scene interaction via prompted chain-of-contacts. In *International Conference on Learning Representations*, 2024. 2

[38] Jiahao Xu, Xiaohan Yuan, Xingchen Wu, Chongyang Xu, Kun Li, and Buzhen Huang. Stability-driven motion generation for object-guided human-human co-manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 38355–38365, 2026. 2, 3

[39] Liang Xu, Yizhou Zhou, Yichao Yan, Xin Jin, Wenhan Zhu, Fengyun Rao, Xiaokang Yang, and Wenjun Zeng. ReGenNet: Towards human action-reaction synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1759–1769, 2024. 3

[40] Sirui Xu, Hung Yu Ling, Yu-Xiong Wang, and Liang-Yan Gui. Intermimic: Towards universal whole-body control for physics-based human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12266–12277, 2025. 2

[41] Sirui Xu, Samuel Schulter, Morteza Ziyadi, Xialin He, Xiaohan Fei, Yu-Xiong Wang, and Liang-Yan Gui. InterPrior: Scaling generative control for physics-based human-object interactions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23516–23527, 2026. 2, 3

[42] Wei Yao, Haohan Ma, Hongwen Zhang, Yunlian Sun, Liangjun Xing, Zhile Yang, Yuanjun Guo, Yebin Liu, and Jinhui Tang. Synagent: Generalizable cooperative humanoid manipulation via solo-to-cooperative agent synergy. *arXiv preprint arXiv:2604.18557*, 2026. 2, 3

[43] Chengxuan Ying, Tianle Cai, Shengjie Luo, Shuxin Zheng, Guolin Ke, Di He, Yanming Shen, and Tie-Yan Liu. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 28877–28888, 2021. 2

[44] Runyi Yu, Yinhuai Wang, Qihan Zhao, Hok Wai Tsui, Jingbo Wang, Ping Tan, and Qifeng Chen. SkillMimic-V2: Learning robust and generalizable interaction skills from sparse and noisy demonstrations. In *ACM SIGGRAPH 2025 Conference Papers*, pages 1–11, 2025. 2

[45] Yunbo Zhang, Deepak Gopinath, Yuting Ye, Jessica Hodgins, Greg Turk, and Jungdam Won. Simulation and retargeting of complex multi-character interactions. In *ACM SIGGRAPH 2023 Conference Proceedings*, pages 1–11, 2023. 2, 3